

Master Index

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Comprehensive, first-principles, interview-focused notes across 11 core SDE/Senior-SDE topics.

Each topic follows the same structure: **Overview** → **Why it exists** → **Why FAANG cares** → **Core concepts** → **Architecture/diagrams** → **Real-world examples** → **Real-life analogy** → **Memory tricks** → **Common interview questions** → **Senior-level discussion** → **Interview tips** → **Revision cheat sheet**.

1 Night-Before Cram

[CHEATSHEET.md](#) — one-page all-topics recall (all 11 topics condensed, ~20 min read). Start here the night before.

2 Magazine Editions (premium PDFs)

Publication-quality, magazine-style PDFs of every topic — custom typography, framed diagrams, infographic callouts and color-coded sections. Each is self-contained (fonts embedded).

Topic	PDF	Topic	PDF
Concurrency	PDF	Distributed Systems	PDF
Operating Systems	PDF	Databases	PDF
Computer Networks	PDF	Performance Eng.	PDF
Cloud & Infra	PDF	Security	PDF
System Design	PDF	Behavioral	PDF
Low-Level Design	PDF		

= flagship hand-crafted edition (custom infographics). Others are generated from the source notes with the same design system. The cheat sheet also has a [print PDF](#). Builders live in `.magazine/`.

Applied-pack magazines:

Pack	PDF	Pack	PDF
DSA · Coding Patterns	PDF	System Design Problems I	PDF
DSA · Data Structures	PDF	System Design Problems II	PDF
DSA · Algorithms	PDF	LLD · Machine Coding	PDF

** AI / ML series** (see [AI-ML/00-INDEX.md](#)):

Edition	PDF	Edition	PDF
ML Fundamentals	PDF	Transformers & LLMs	PDF
Classic ML Algorithms	PDF	LLM Applications & GenAI	PDF
Deep Learning	PDF	MLOps & ML System Design	PDF

Preview & edit any edition live: `cd .magazine && python3 serve.py` → <http://localhost:8000>

3 Topics

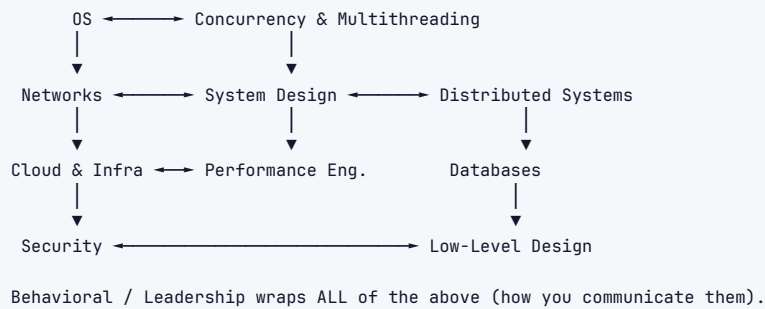
#	Topic	File
1	Concurrency & Multithreading	01-concurrency-multithreading.md
2	Operating Systems	02-operating-systems.md
3	Computer Networks	03-computer-networks.md
4	Cloud & Infrastructure	04-cloud-infrastructure.md
5	System Design	05-system-design.md
6	Low-Level Design (LLD)	06-low-level-design.md
6½	Object-Oriented Programming (OOP)	12-oop-concepts.md
7	Distributed Systems	07-distributed-systems.md
8	Databases	08-databases.md
9	Performance Engineering	09-performance-engineering.md
10	Security Fundamentals	10-security-fundamentals.md
11	Behavioral / Leadership / Googleyness	11-behavioral-leadership.md
13	Design Patterns	13-design-patterns.md
14	API Design	14-api-design.md
15	Caching	15-caching.md
16	Message Queues & Streaming	16-message-queues.md
17	Testing & Code Quality	17-testing.md
18	Observability	18-observability.md
19	Git & Dev Workflow	19-git-version-control.md
20	Linux & the Command Line	20-linux-cli.md
21	Vector Databases & Embeddings	21-vector-databases.md

AI Agents & Tool Use lives in the [AI-ML series](#).

4 Applied Practice Packs

Pack	What's inside
DSA / Coding Round	Pattern recognition + templates, data structures, algorithms & complexity — the #1 interview filter.
System Design Problems	8 fully worked end-to-end designs: Part 1 — TinyURL, Rate Limiter, News Feed, Chat · Part 2 — YouTube, Uber, Distributed Cache, Web Crawler
LLD / Machine-Coding	6 complete runnable Python solutions: LRU Cache, Parking Lot, Rate Limiter, Splitwise, Elevator, Tic-Tac-Toe.

5 How topics connect



6 Suggested revision order (last-week-before-interview)

1. System Design (5) + Distributed Systems (7) – highest signal for senior loops
2. Databases (8) + Concurrency (1) – deep technical follow-ups
3. Networks (3) + OS (2) – fundamentals screens
4. LLD (6) + Cloud (4) + Performance (9) + Security (10)
5. Behavioral (11) – review your STAR stories nightly